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DATA LITERACY

ADRP-SL

USAREUR-AF

DATA LITERACY FOR SENIOR LEADERS

COMMANDING GENERAL'S NOTE

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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DATA LITERACY FOR SENIOR LEADERS

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NOTE

This publication is the senior leader companion to Data Literacy Technical Reference, Data Literacy. It presents foundational principles and command responsibilities without procedural detail. Commanders and senior staff should read this publication; those requiring technical depth should consult Data Literacy Technical Reference.

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[INSERT: General Donahue's signed data guidance — to be provided by G6/Data and inserted here prior to publication.]

PREFACE

This publication establishes principles and command responsibilities for data literacy across United States Army Europe and Africa (USAREUR-AF). It is written for O-5 and above, Command Sergeant Major (CSM) and Sergeant Major (SGM), and senior civilian equivalents.

USAREUR-AF is the Army Service Component Command (ASCC) to United States European Command (USEUCOM), responsible for theater land operations from Iceland to the Caucasus. Subordinate commands include V Corps (Forward), 21st Theater Sustainment Command (21 TSC), and 7th Army

Training Command (7ATC). USAREUR-AF supports NATO Article 5 collective defense commitments and operates in a coalition environment requiring data interoperability with Allied nations. These realities make data literacy more operationally consequential here than in any peacetime garrison environment. USAREUR-AF formations operate in a contested information environment in which adversary forces actively exploit data gaps and analytical failures.

This publication implements ADP 3-13, Information, by establishing the data literacy foundation required for information advantage in the USAREUR-AF AOR. ADP 3-13 defines information as combat power. Data literacy is what makes that power accessible to commanders. This publication tells senior leaders what they are responsible for and why it matters in the context of that broader doctrine.

Data literacy is a command competency. It is not a technical skill reserved for analysts. Commanders who understand how to direct, evaluate, and act on data products will make faster, better decisions than those who do not. Those who cannot are vulnerable to surprise, manipulation, and misinformed action.

This publication does not contain procedures. It contains principles and responsibilities. Technical guidance belongs in the Data Literacy Technical Reference. This document tells senior leaders what they are responsible for and why it matters.

CHAPTER 1 — WHY DATA LITERACY MATTERS FOR SENIOR LEADERS

BLUF: Data-literate commands win. Data-illiterate commands are surprised. The commander sets the standard.

1-1. Bottom Line Up Front

Every modern military operation generates and depends on data. Logistics status, personnel readiness, threat assessments, route conditions, and sensor feeds are all data products. Commanders who understand what data tells them — and what it does not — make better decisions under pressure. Commanders who cannot read, question, or direct data products cede decision advantage to the enemy, to their own biases, and to whatever their staff decides to show them.

Data literacy for senior leaders is a combat multiplier.

1-2. The Operational Reality

1-2. The Commanding General of USAREUR-AF has stated directly: **"The race to harness live data will determine who wins the next war."** This is not a warning about future conflict. It is a description of the competition underway now. In Europe, USAREUR-AF and NATO ground forces are already integrating allied sensors, data, and AI-powered machine learning tools into real-time decision-making. As the CG

has noted: **"This is not a projected capability or an operational needs statement to inform future investment. This is what we are doing today."** The implications for command are immediate. Formations whose leaders cannot direct, evaluate, and integrate data products are already behind.

1-3. Commanders already depend on data, whether they recognize it or not. Every Common Operating Picture (COP) update is a data product. Every readiness report is a data product. Every intelligence summary is built from raw data that has been collected, processed, and assessed. The question is not whether commanders use data — they do. The question is whether they use it deliberately, critically, and effectively.

1-4. In USAREUR-AF's operational environment, the volume of data available to commanders has increased faster than formations' capacity to use it. Sensors, digital sustainment systems, personnel databases, and joint feeds produce more information than any individual can process. Without data-literate leaders directing the analytical effort, formations either drown in data or ignore it entirely. Both outcomes degrade decision quality.

1-5. The commander's data challenge is not access — it is direction, evaluation, and integration. A formation that can collect data but cannot interpret it honestly, question its reliability, or integrate it into planning has not gained an advantage. It has added noise.

NOTE

The command-and-control system USAREUR-AF has built in Europe represents a level of transformation comparable to radio-enabled command in World War II. The CG has directed that this system be rapidly scaled and connected globally. Data-literate leaders at every echelon are required to sustain and extend that advantage.

1-3. What Data Literacy Means for a Senior Leader

1-5. Data literacy for a senior leader is NOT the ability to write code, build databases, or design analytical pipelines. Those are technical skills. Senior leaders require a different competency.

1-6. A data-literate senior leader can do the following:

Competency	Description
Direct	Specify what data is needed and why, linked to Commander's Critical Information Requirements (CCIR)
Question	Ask the right questions about a data product before accepting its conclusions
Evaluate	Identify red flags — gaps, inconsistencies, unsupported claims — in data products
Integrate	Connect data products to decisions without over-relying on any single source
Resource	Allocate people, time, and tools to data tasks as deliberate capability investment
Protect	Enforce classification and access control discipline as a command responsibility

1-7. These competencies are exercised in the same way commanders exercise all other leadership skills: through deliberate practice, by demanding standards, and by modeling the behavior they expect from their subordinates.

1-4. The VAUTI Standard — What Makes Data Trustworthy

1-8. Not all data is equally useful. Before acting on a data product, commanders and staff must assess it against five criteria derived from the DoD Data Strategy (2020): Visible, Accessible, Understandable, Trustworthy, and Interoperable — the VAUTI standard.

Criterion	Meaning	Commander's Question
Visible	Data is discoverable — personnel know it exists	Can we find this data when we need it?
Accessible	Data can be retrieved by those who need it	Do the right people have access?
Understandable	Data meaning, context, and limitations are clear	Does the staff know what this actually measures?
Trustworthy	Data is accurate, current, and from a verified source	Do we have confidence in the source and method?
Interoperable	Data can be shared and used across systems and organizations	Can we exchange this with joint and Allied partners?

1-9. VAUTI is not a checklist — it is a decision framework. A data product that fails any VAUTI criterion introduces risk into the decision it supports. Commanders must require that data products used for significant decisions are explicitly evaluated against all five criteria.

1-10. In the USEUCOM AOR, interoperability carries additional weight. Data products shared with NATO Allies or partner nations must meet both U.S. standards and applicable NATO data exchange standards. A product that is trustworthy within a U.S. system but not interoperable outside it limits coalition effectiveness. The CG USAREUR-AF has identified interoperability as Allied nations' **"number one**

concern" and has been direct about the root cause: **"It's trust. That's all that is. We can solve it."** Data standards, disciplined handling, and rigorous security protocols are how commanders build that trust at the working level.

1-5. Decision Dominance

1-11. Decision dominance is the operational objective of superior data and analytical capability: to make better decisions than the adversary, faster, and at higher confidence. It is not a technological state — it is a command state achieved through disciplined data collection, rigorous analysis, and effective integration of information into command decisions.

1-12. Decision dominance is an enduring objective regardless of the operational environment. In the USAREUR-AF AOR, where adversary forces invest heavily in deception operations, electronic warfare, and information operations designed to degrade Allied situational awareness, decision dominance is a competitive requirement. USAREUR-AF formations that consistently produce timely, high-quality data products — and whose commanders know how to use them — maintain the operational initiative. Those that do not cede it.

1-13. Senior leaders are responsible for building and maintaining their formation's capacity for decision dominance. This means investing in data capability as a readiness priority — not as an administrative task.

1-6. The Cost of Data Illiteracy — Three Vignettes

Vignette 1 — The Invisible Readiness Gap

1-14. A brigade commander reviewed weekly readiness reports and consistently saw green across all subordinate units. Three days before a V Corps exercise at Grafenwöhr, the brigade discovered that one battalion had been manually entering estimated values for deadline equipment rather than pulling live data from the maintenance system. The exercise launched with a shortfall of 22 percent deadlined vehicles that no brigade-level leader had known about. The commander had never asked how the data was collected or validated.

Vignette 2 — The Confident Wrong Answer

1-15. A staff officer presented a route analysis showing three viable MSRs (Main Supply Routes) into an assembly area near the Suwałki Gap, supported by a detailed chart with color coding and statistics. The commander approved the plan. During execution, the lead element encountered an impassable bridge — a condition documented in an engineer survey from the previous week that had not been integrated into the analysis. The data product had looked authoritative. The commander had not asked what data sources it drew from or how current they were.

Vignette 3 — The Culture That Punished Honesty

1-16. A battalion intelligence section consistently reported high-confidence threat assessments. Pressure from senior leaders for clear, actionable products had led analysts to remove uncertainty language from their reports. The battalion operated for weeks with misplaced confidence in threat assessments that had significant gaps. When contact occurred in an unexpected location along a Baltic flank corridor, the commander was surprised. The staff had known about the gaps. They had learned not to report them.

1-7. The Commander's Role in Building a Data-Literate Formation

1-17. Culture is set by command. If the commander does not ask data-driven questions, subordinates will not prepare data-driven answers. If the commander accepts charts without questioning their sources, staff will produce charts without validating their sources. The standard the commander walks past is the standard the formation adopts.

1-18. Building a data-literate formation requires three things from senior leaders. First, demand honest data products — including uncertainty. Second, resource the capability to produce them. Third, model data-critical behavior in every briefing, planning session, and after-action review (AAR).

1-19. This is not an additional task. It is the commander's job, applied to the information domain.

1-8. Doctrinal Foundation — ADP 3-13 and Information Advantage

SIDEBAR: ADP 3-13 — Data Is the Raw Material of Combat Power

ADP 3-13, *Information* (2023), establishes that information is combat power. It defines *information advantage* as the operational condition achieved when a force generates, protects, and exploits information more effectively than the adversary.

Data literacy is the prerequisite for that advantage.

Information advantage begins with data. Without disciplined data collection, rigorous data quality, and leaders who can direct and evaluate data products, no formation can generate the analytical outputs that information advantage requires. A formation with excellent data practices is more resistant to adversary deception, better able to maintain situational awareness under pressure, and faster to recognize and act on opportunity.

For USAREUR-AF, operating within a coalition theater where adversaries actively contest the information environment, this connection is operationally urgent. Building data literacy is not separate from information operations — it is its foundation.

Senior leaders who invest in their formation's data capability are building information power as defined in ADP 3-13.

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1-20. This publication implements ADP 3-13, Information, by establishing the data literacy foundation required for information advantage in the USAREUR-AF AOR. ADP 3-13 establishes that information is combat power — that commanders who can generate, protect, and exploit information more effectively than the adversary gain a decisive operational advantage.

1-21. Data literacy is not separate from information operations. It is its prerequisite. A commander who cannot direct and evaluate data products cannot generate the quality analytical outputs that information advantage requires. A formation that produces poor-quality data cannot supply the situational awareness that information advantage depends on. Building data literacy in a formation is, therefore, directly building the command's capacity for information operations as defined in ADP 3-13.

1-22. In the USAREUR-AF AOR, this connection is operationally urgent. USAREUR-AF operates in a contested information environment across the USEUCOM theater. Adversary forces invest heavily in deception, electronic warfare, and information operations precisely to degrade Allied situational awareness. USAREUR-AF formations that produce rigorous, sourced, timely data products are more resistant to that adversary effort than those that do not.

1-9. NATO and EUCOM AOR Context

1-23. USAREUR-AF data operations must comply with NATO Architecture Framework version 4 (NAFv4) and support interoperability with Allied Joint Publications (AJP-01, AJP-3, AJP-5). In practical terms, this means that data systems, data products, and data sharing practices must be designed for coalition use — not only for unilateral Army operations. Commanders must ensure that data products created for sharing with Allied or partner nation forces are appropriately marked under both U.S. and NATO classification standards and that data systems can exchange information with Allied equivalents.

1-24. The CG USAREUR-AF has described Europe as the **"premier battle laboratory"** for building the lethal, data-driven force required to meet any adversary. The real-time command and control system established in USAREUR-AF — integrating Allied sensors, data, and AI-powered analytical tools — must now be scaled globally. USAREUR-AF commanders and formations that master data-driven operations in this theater are building the template the Army will export worldwide.

1-25. Data literacy requirements in this publication and in Data Literacy Technical Reference apply equally to multinational operations and to unilateral Army operations. The analytical standards, quality requirements, and security disciplines described here are not reduced in coalition contexts — they are essential to coalition effectiveness.

CHAPTER 2 — CORE PRINCIPLES OF DATA-DRIVEN COMMAND

BLUF: Eight principles define how commanders should think about and use data. These principles apply regardless of the type of operation, the tool, or the echelon.

2-1. Principle 1 — Data Is Ammunition

2-1. Data is a tactical and operational resource. Like ammunition, it must be collected deliberately, protected from the adversary, and expended at the right time for the right purpose. A formation that collects data carelessly, shares it without access controls, or wastes analytical capacity on low-priority questions is mismanaging a key resource. Commanders must treat data requirements with the same rigor they apply to logistics planning.

2-2. Principle 2 — Quality Over Quantity

2-2. One trustworthy data product is worth more than ten unreliable ones. Formations under time pressure will default to the most available data, even if it is poor quality. Commanders must establish and enforce data quality standards before operations, not during them. Accepting degraded data under pressure normalizes the practice. When quality is uncertain, commanders must demand explicit uncertainty statements rather than false precision.

2-3. Principle 3 — Context Converts Data Into Intelligence and Decisions

2-3. Raw data has no value without context. A number means nothing without knowing what it measures, when it was collected, how it was collected, and what it is being compared to. A rate of change means nothing without knowing the baseline. Commanders must demand that data products include context — source, collection method, time, and known limitations — not just findings. Analysis without context is assertion.

2-4. Principle 4 — Commanders Must Ask the Right Questions

2-4. Analytical effort follows commander guidance. If the commander does not specify what questions need answers, analysts will answer the questions they can answer — which may not be the questions that matter. Establishing Priority Intelligence Requirements (PIR) and Friendly Force Information Requirements (FFIR) is the mechanism by which commanders direct analytical effort. A vague CCIR produces vague data products. A precise requirement produces focused, actionable analysis.

2-5. Principle 5 — Automation Enables Scale; Judgment Enables Decisions

2-5. Automated systems can process data at volumes and speeds no human can match. They can flag anomalies, generate reports, and aggregate feeds from across the formation. They cannot exercise judgment. Commanders must understand what their automated systems are and are not doing. Automation does not remove the human from the decision. It informs the human. Commanders who treat automated outputs as decisions — rather than inputs — have surrendered their judgment to their tools.

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2-6. Principle 6 — Data Lineage Enables Trust and Accountability

2-6. Data lineage is the documented trail of where data came from, who handled it, what transformations were applied, and when. Without lineage, a commander cannot assess whether a data product is trustworthy. Without accountability for data — who collected it, who processed it, who approved it — errors cannot be corrected and bad practices cannot be identified. Commanders must require that data products used for significant decisions include a clear lineage statement.

2-7. Principle 7 — Security Is Not Optional

2-7. Data classification and access control are command responsibilities. Operational data, personnel data, intelligence feeds, and logistics systems all carry classification requirements. Data aggregation — combining multiple unclassified elements — can produce a classified product. Commanders must enforce classification discipline and must ensure that data sharing practices within and outside the formation comply with applicable security requirements. A data breach caused by poor access discipline is a command failure.

2-8. Principle 8 — Culture Is Set by Command

2-8. If the commander does not use data, neither will the formation. If the commander accepts unsupported assertions in briefings, staff will produce unsupported assertions. If the commander punishes uncertainty — by reacting negatively to "we don't know" answers — the formation will stop reporting uncertainty and start producing false confidence. The commander's daily behavior in briefings, planning sessions, and AARs defines the formation's data culture. No training program can compensate for a command climate that does not value honest, rigorous data use.

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CHAPTER 3 — COMMANDER'S DATA RESPONSIBILITIES

BLUF: Commanders have seven core data responsibilities. These are not delegated tasks — they are command functions that require commander attention, direction, and enforcement. The C2DAO is the commander's primary advisor and enforcer across all of them.

3-1. The Command Chief Data and Analytics Officer (C2DAO)

3-1. USAREUR-AF operates a Command Chief Data and Analytics Officer (C2DAO). The C2DAO is the commander's primary advisor on data governance, data quality, platform use, and compliance with Army enterprise data policy. This officer does not create Army enterprise data policy — that authority resides at HQDA. The C2DAO consumes, implements, and enforces that policy at the command level.

3-2. The chain of authority for data governance flows as follows:

Echelon	Role
HQDA MADOs (Major Army Data Owners)	Establish enterprise data standards and policy
Data Stewards	Own and enforce standards for designated data domains
Functional Data Managers	Manage data within functional areas (personnel, logistics, intelligence)
C2DAO (Command Level)	Advise the commander; integrate governance across functions; enforce compliance

3-3. The C2DAO is effective only to the extent the commander empowers the role. A C2DAO without resourcing, access to leadership, and command backing cannot enforce standards across a formation. Commanders are responsible for resourcing and empowering their C2DAO. Failure to do so is a command decision, and the formation's data quality will reflect it.

3-4. Commanders should use the C2DAO as a trusted advisor in the following areas: - Assessing the formation's data governance posture. - Evaluating readiness of data systems before major operations or exercises. - Reviewing compliance with Army CIO guidance (2024) and the Unified Data Reference Architecture (UDRA) v1.1 (2025). - Advising on data sharing arrangements with NATO Allies and partner nations. - Escalating data quality or security issues that cross functional boundaries.

3-5. The C2DAO role is not a technical position in isolation. It is a command-level governance function. Commanders who treat it as an administrative billet misunderstand its operational value.

3-2. Establishing Data Requirements

3-6. Data requirements must be linked to decisions. Every CCIR — whether a PIR or FFIR — implies a data requirement. Commanders must be able to answer: what data will tell me whether this requirement has been satisfied? Vague CCIRs produce vague data collection and analysis. Precise requirements produce focused, actionable products.

3-7. Commanders establish data requirements during mission analysis and refine them throughout the Military Decision-Making Process (MDMP). Data requirements should be reviewed when the operational situation changes significantly enough to alter what information matters most. Stale requirements produce stale data effort.

3-8. The following questions help commanders establish effective data requirements:

Question	Purpose
What decision does this data support?	Ensures data serves a decision, not a briefing
What does "satisfied" look like?	Creates a clear threshold for action
When do I need it?	Sets a collection and reporting timeline
What is the acceptable confidence level?	Defines quality standard up front
What happens if I don't get it?	Forces assessment of the decision risk

3-3. Appointing and Supporting Data Stewards

3-9. Every formation that manages significant data — whether personnel records, logistics status, intelligence products, or training databases — needs a data steward. A data steward is the individual responsible for the accuracy, completeness, and appropriate handling of a specific dataset or system.

3-10. Commanders must formally appoint data stewards and define their responsibilities in writing. Appointment without authority is ineffective. Data stewards must have the authority to enforce data entry standards, flag inaccuracies to leadership, and decline to sign off on products that do not meet quality standards.

3-11. Data stewards are not administrators. They are quality owners. Commanders who treat the data steward role as an additional duty managed at the lowest possible rank will get the data quality that posture deserves.

3-4. Demanding Data Quality Standards

3-12. Commanders define quality by what they accept. A formation that routinely produces data products with undocumented sources, outdated figures, or missing uncertainty statements will continue to do so until the commander stops accepting them.

3-13. Commanders should apply the following minimum standards when receiving data products in any formal setting:

- Source and collection method are identified.
- Data currency (age) is stated.
- Known gaps or limitations are disclosed.
- Confidence levels are expressed — not implied.
- Derived figures (calculations, estimates) are distinguished from raw collected data.

3-14. These standards should be enforced consistently, not just in high-stakes briefings. The formation learns from everyday practice. Inconsistent standards produce inconsistent quality.

3-5. Data in the MDMP

3-15. Data requirements are established and refined at each step of the MDMP. The commander's role at each step is distinct:

MDMP Step	Commander's Data Role
Receipt of Mission	Identify initial information gaps; issue initial CCIR
Mission Analysis	Validate data requirements against mission variables; assign collection tasks
COA Development	Specify what data each Course of Action (COA) assumes; flag untested assumptions
COA Analysis (Wargame)	Test COAs against data uncertainty — what if the data is wrong?
COA Comparison	Require explicit data basis for each comparative factor
Orders Production	Ensure data requirements and reporting triggers are included in the OPORD
Execution and Assessment	Monitor CCIR status; adjust collection effort as the situation develops

3-16. Data literacy in MDMP is not a new step. It is the habit of treating information products with the same rigor applied to other planning products — and holding the staff accountable for the same standards.

3-6. Resourcing Data Capability

3-17. Data capability does not self-generate. It requires people with skills, tools that work, time to do the analysis, and a command climate that rewards doing it well. Commanders must resource data capability deliberately, as they resource any other operational function.

3-18. Commanders should assess their formation's data capability against three factors:

People. Do staff sections have personnel with sufficient skill to collect, manage, and analyze operational data? Are those personnel employed on data tasks or diverted to other duties? Is there a training path that develops data skills over time?

Tools. Do the tools available to the formation meet its data needs? Are they operational, licensed, and maintained? Do personnel know how to use them effectively?

Time. Is analytical effort protected from being consumed by administrative tasks? Is there time in the battle rhythm for analysis — not just reporting? Are deadlines realistic given the available data and analytical capacity?

3-19. Underfunding data capability while demanding data products is a command contradiction. Commanders must close the gap between what they ask for and what they resource.

3-7. Enforcing Data Security and Classification Discipline

3-20. Commanders are responsible for the classification and access control practices of their formation. This includes digital data — files, databases, reports, and shared drives — as well as physical products.

3-21. Commanders must ensure their formations observe the following:

- Data products are classified at the appropriate level based on content, including aggregation effects.
- Access to sensitive data systems is granted on a need-to-know basis, not convenience.
- Personnel transfers and departures are accompanied by prompt access revocation.
- Data sharing with partner nations, joint forces, or external agencies follows applicable security agreements.
- Data breaches or suspected unauthorized access are reported immediately through proper channels.

3-22. In the USEUCOM AOR, coalition data sharing carries additional complexity. Data products shared with NATO Allies must comply with both U.S. classification standards and applicable NATO handling requirements. Commanders must ensure their C2DAO and security staff are synchronized on this requirement before any data exchange arrangement is established with a partner nation.

3-23. Classification discipline is not the S-6's problem. It is a command problem. Commanders who treat it as a technical issue rather than a command responsibility will eventually own the consequences of that posture.

CHAPTER 4 — EVALUATING DATA PRODUCTS

BLUF: Commanders must be active consumers of data products — not passive recipients. Every briefing is an opportunity to enforce standards, identify gaps, and direct the analytical effort.

4-1. Questions Every Commander Should Ask

4-1. When presented with a data product — a chart, a report, an assessment, a dashboard — commanders should ask a consistent set of questions before accepting its conclusions. These questions are not expressions of distrust. They are the minimum standard for informed decision-making.

Question	What It Uncovers
What is the source?	Identifies the origin and potential bias of the data
How was this collected?	Reveals method limitations, sampling gaps, or reporting errors
How current is this data?	Flags whether conditions may have changed
What is not included here?	Exposes gaps the product may not advertise
What is the confidence level?	Distinguishes high-quality findings from estimates
What assumption is this built on?	Identifies untested premises that may invalidate the product
What would change this conclusion?	Tests the robustness of the finding
Who produced this and do they have conflicts of interest?	Flags organizational or analytical bias

4-2. These questions take seconds to ask. They create a command culture that demands accountability from analysts and protects commanders from acting on flawed products.

4-2. Red Flags in Data Products

4-3. Certain characteristics of a data product should prompt the commander to pause and dig deeper before acting on its conclusions. These are not disqualifying — but they require explanation.

Red Flag 1 — No Uncertainty Stated. Any data product that presents findings without confidence levels, margins, or acknowledgment of limitations is incomplete. Real analysis always has uncertainty. Products with none have had it removed — which is worse than products that never measured it.

Red Flag 2 — Precision Without Basis. A report that states "enemy force is 847 personnel" when the collection method is observer reports and imagery should be questioned. False precision signals that numbers have been manipulated to appear more authoritative than the underlying data supports.

Red Flag 3 — Conclusions That Confirm Existing Assumptions. If every data product confirms what the commander and staff already believe, the analytical process may be producing advocacy rather than analysis. Honest data frequently surprises. Products that never do may have been shaped to avoid discomfort.

Red Flag 4 — Missing Data Described as Zero. "No reports of enemy activity in sector" is not the same as "no enemy activity in sector." Missing data should be disclosed as missing, not treated as evidence of absence.

Red Flag 5 — No Audit Trail. If the analyst cannot explain where the data came from, what transformations were applied, and who reviewed it, the product does not meet minimum standards for command-level decisions.

Red Flag 6 — Single-Source Products for High-Stakes Decisions. One source — regardless of confidence level — is insufficient for decisions with significant operational consequences. Commanders should require corroboration before acting on single-source products.

4-3. Understanding Uncertainty and Confidence Levels

4-4. Uncertainty is not weakness. It is honesty. Commanders who demand certainty from analysts will receive false certainty. Commanders who accept and act on honest uncertainty will make better decisions than those who require false confidence before they can act.

4-5. Confidence levels in analytical products typically reflect two factors: the quality of the underlying data and the strength of the reasoning applied to it. A high-confidence assessment based on poor data is not trustworthy. A low-confidence assessment based on strong data may be more reliable than it appears. Commanders must understand both factors.

4-6. The appropriate response to uncertainty is not paralysis — it is risk management. Commanders identify what they know, what they do not know, and what they need to find out before the decision point. They assign collection tasks to close critical gaps and make decisions with explicit acknowledgment of residual uncertainty.

4-7. NOTE: Intelligence products use defined confidence level terminology (high, moderate, low). When working with non-intelligence data products — logistics, personnel, engineering — commanders should require equivalent language. "We think" and "the data suggests" are not confidence statements. "Based on three independent sources with collection from the last 48 hours, we assess with moderate confidence" is.

4-4. Distinguishing Correlation from Causation

4-8. Correlation means two things move together. Causation means one thing causes another. Confusing them is one of the most common analytical errors in briefings, and it leads commanders to take actions that do not produce expected results.

4-9. Example: A unit observes that incidents increase on days when a particular patrol route is used along a contested corridor in the eastern AOR. That is correlation. The cause may be the route — or the cause may be that the route is used when threat conditions are already elevated. Acting on correlation as if it were causation (stopping the route) may not reduce incidents if the route was not the cause.

4-10. Commanders should require analysts to explicitly state whether a relationship is correlational or causal, and to explain the basis for that claim. The question "how do we know this causes that, rather than just being associated with it?" is always appropriate.

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4-5. Directing the Analytical Effort

4-11. Commanders direct analytical effort through their requirements, not through telling analysts what conclusions to reach. The distinction is critical. Directing requirements preserves analytical integrity. Directing conclusions corrupts it.

4-12. Commanders should specify: - What question needs to be answered. - What decision the answer will inform. - When the answer is needed. - What level of confidence is acceptable given the stakes.

4-13. Commanders should not specify: - What answer is expected. - What data sources to prioritize in ways that bias the outcome. - That uncertainty should be removed from products before briefing.

4-14. Analysts who are directed toward conclusions rather than questions will eventually produce the conclusions the commander wants — regardless of what the data says. That is how formations end up surprised.

CHAPTER 5 — BUILDING A DATA-LITERATE FORMATION

BLUF: Data literacy is a trained and developed competency, not a natural talent. Commanders build it through deliberate training, integration into collective events, and consistent command climate.

5-1. Training Requirements and the Data Literacy Continuum

5-1. Data literacy exists on a continuum. Not every Soldier needs the same level of proficiency. What matters is that each role has a defined standard, that training is available to meet it, and that leaders enforce the standard.

5-2. The USAREUR-AF data literacy training framework consists of five tiers, organized as Technical Manuals (TMs). The first three tiers apply to all personnel; the -40 and -50 tiers are specialist tracks for designated data roles.

Tier	Target Audience	Core Competency
TM-10 (Operator)	All personnel — all MOS/staff	Navigate and consume data products on Maven Smart System; observe data security and accuracy standards
TM-20 (Builder)	All personnel — builds on TM-10	Build basic Workshop applications; run light transforms; manipulate and share data products without specialist tools
TM-30 (Advanced Builder)	Data-adjacent personnel (17/25-series, S6/G6/G2/G9)	Design pipelines and Ontology objects; build complex dashboards; apply governed data patterns
TM-40 Series (Specialist)	Designated specialist roles	Role-specific mastery: ORSA (40A), AI Engineer (40B), ML Engineer (40C), Program Manager (40D), Knowledge Manager (40E), Software Engineer (40F)
TM-50 Series (Advanced Specialist)	TM-40 graduates in senior/lead roles	Advanced practitioner: independent research-grade capability in each specialist track; leads and mentors others

5-3. Senior leaders sit at the TM-10 level for technical skills and at a distinct "Commander" level for direction, evaluation, and command responsibilities. This publication describes the Commander level. Data Literacy Technical Reference describes all levels in detail. TM-40 and TM-50 track training is documented in the corresponding Technical Manuals.

5-2. Summary of Training Levels

5-4. All training levels are cumulative. TM-20 competencies build on TM-10; TM-30 builds on TM-20; the TM-40/50 specialist tracks require TM-30 as a prerequisite. Commanders must ensure their formation has personnel trained to the appropriate tier for each data function in use.

5-5. NOTE: Gaps in training coverage are a readiness issue, not an administrative issue. A formation with no personnel trained to TM-20 on a critical data system is as vulnerable as a formation with no personnel qualified on a critical weapons system. Treat it accordingly.

5-6. Key training outcomes by tier:

TM-10 — Operator (All Personnel) - Navigate Maven Smart System and access relevant data products. - Understand the difference between data and information. - Know responsibilities for accurate data entry. - Recognize and report data quality problems. - Observe classification and access control requirements.

TM-20 — Builder (All Personnel) - Build basic Workshop applications and simple pipelines. - Apply data quality standards to collection and reporting. - Identify and escalate data integrity issues. - Produce basic analytical products from available tools.

TM-30 — Advanced Builder (Data-Adjacent Personnel) - Design Foundry pipelines and Ontology objects. - Build governed data products for use across the formation. - Perform analytical validation and data profiling. - Support the data requirements process and mentor TM-10/20 users.

TM-40 Series — Specialist (Designated Roles) - Achieve role-specific mastery within a defined track (ORSA, AI Eng, ML Eng, Program Manager, Knowledge Manager, Software Engineer). - Produce command-level data products independently. - Apply advanced analytical, engineering, or operational methods within the track.

TM-50 Series — Advanced Specialist (Senior/Lead Roles) - Lead and mentor within the specialist track. - Conduct research-grade analysis and independently solve novel problems. - Design new frameworks, tools, and capability for the formation. - Develop and update training materials for the track.

5-3. Integrating Data Tasks Into Collective Training

5-7. Data literacy is best developed in context — during the tasks it supports. Training data skills in isolation from operational context produces personnel who understand concepts but cannot apply them under pressure.

5-8. Commanders must integrate data tasks into the following collective training events:

Situational Training Exercises (STX). Include data entry, reporting, and quality check tasks in STX lane design. Train Soldiers and NCOs to enter accurate data under time pressure and to flag discrepancies.

Field Training Exercises (FTX). Operate the formation's data systems in field conditions. Require data products — readiness reports, logistics status, assessment products — to meet quality standards throughout the exercise. Use AARs to identify and correct data discipline failures.

Command Post Exercises (CPX). Exercise the full data product chain — from collection requirement to command-level brief. Require staff to source their briefing products and disclose uncertainty. Practice the commander's questioning routine on every data product presented.

5-9. NOTE: If data quality standards are relaxed during training because of time pressure or administrative convenience, the formation will relax them during operations for the same reasons. Train the standard you intend to enforce.

5-4. Creating Conditions for a Data-Driven Culture

5-10. Culture is shaped by what commanders recognize, reward, and require. To build a data-driven culture, commanders must make all three visible and consistent.

Recognition. Identify and recognize personnel who catch data errors, produce high-quality analytical products, or improve data practices in their section. Make it visible that the command values data quality — not just data volume.

Expectation. Set explicit standards for data products in all formal briefings and planning events. Publish those standards. Review them at leader professional development (LPD) sessions. Apply them consistently, not only when the stakes are obvious.

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Resourcing. Fund training time, protect analytical personnel from unrelated taskings, and acquire tools that enable the standard. A command that requires excellent data products but does not provide time, tools, or training has made an unfunded mandate — and the staff knows it.

5-11. Senior NCO leadership is critical. CSMs and SGMs who enforce data discipline in unit administration, training management, and NCO development programs reinforce the command climate that the commander sets at the top.

5-5. Key Indicators of a Data-Literate Formation

5-12. Commanders can assess their formation's data literacy through observable indicators. These are not metrics — they are behavioral and cultural signs that the formation has internalized the standard.

Indicator	What It Signals
Staff volunteers uncertainty in briefings without being asked	Formation trusts that honesty is safe
Data products include source, currency, and confidence level by default	Quality standards are internalized, not enforced by exception
Data errors are reported up, not hidden	Command climate rewards accuracy over appearance
CCIRs drive collection planning, not reporting formats	Data requirements are linked to decisions
Data stewards are empowered to reject non-compliant products	Authority matches responsibility
Collective training events use live data systems under realistic conditions	Training replicates operational conditions
After-action reviews identify and correct data failures, not just tactical failures	Data is treated as an operational domain

5-13. A formation that exhibits these indicators has internalized data literacy as a command competency. A formation that exhibits none of them has significant vulnerability in the information domain.

KEY TERMS

The following terms are used throughout this publication. Definitions are provided as they apply to this document. More complete definitions appear in Data Literacy Technical Reference and applicable joint and Army doctrine.

AAR (After-Action Review) — Structured review of events to identify what happened and why.

Analytical Pipeline — Sequence of steps that transform raw data into an analytical product.

Audit Trail — Documentation of data origin, handling, and transformations.

C2DAO (Command Chief Data and Analytics Officer) — The commander's primary advisor on data governance, data quality, platform use, and compliance with Army enterprise data policy at the command level. The C2DAO implements and enforces enterprise policy established by HQDA MADOs; does not create that policy.

CCIR (Commander's Critical Information Requirements) — Information the commander requires to make timely decisions; includes PIR and FFIR.

Confidence Level — Expressed degree of certainty in an analytical assessment.

COP (Common Operating Picture) — Shared display of relevant operational information.

Correlation — Statistical association between two variables; does not imply causation.

Causation — One variable directly produces change in another.

Data Currency — How recent the data is; the age of a data product.

Data Lineage — Documented chain of a dataset's origin, handling, and transformation.

Data Literacy — Ability to read, evaluate, direct, and act on data products at a level appropriate to one's role.

Data Quality — Degree to which data is accurate, complete, consistent, and timely.

Data Steward — Individual responsible for accuracy and appropriate handling of a dataset.

Decision Dominance — The operational state in which a force makes better decisions than the adversary, faster, and at higher confidence — achieved through superior data collection, analytical rigor, and effective command use of information products.

FFIR (Friendly Force Information Requirements) — Information the commander needs about friendly forces to make decisions.

HQDA MADOs (Major Army Data Owners) — HQDA organizations with enterprise authority over designated Army data domains. They establish the standards and policy that Data Stewards, Functional Data Managers, and C2DAOs implement at command level.

MDMP (Military Decision-Making Process) — Army's deliberate planning process; seven steps from receipt of mission through orders production.

PIR (Priority Intelligence Requirements) — Intelligence the commander needs to understand the threat and environment.

Red Flag — Characteristic of a data product that warrants additional scrutiny before use.

UDRA (Unified Data Reference Architecture) — The Army's enterprise data architecture standard. UDRA v1.1 (2025) is the current version codifying data structure and governance requirements across Army systems.

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Uncertainty — Acknowledged limits of confidence in data or analytical conclusions.

VAUTI — The DoD Data Strategy (2020) criteria for trustworthy data: Visible, Accessible, Understandable, Trustworthy, and Interoperable. Used by commanders to assess data product quality before acting on findings.

ADP 3-13 — Army Doctrine Publication 3-13, Information. Establishes information as combat power and defines information advantage as a key operational objective. Data literacy is a prerequisite for information advantage.

Information Advantage — As defined in ADP 3-13, the operational condition achieved when a force can generate, protect, and exploit information more effectively than the adversary.

JADC2 (Joint All-Domain Command and Control) — The DoD concept for connecting sensors, shooters, and decision-makers across all domains. USAREUR-AF is integrated into JADC2.

NAFv4 (NATO Architecture Framework version 4) — The NATO standard for data and system architecture, required for coalition interoperability.

USEUCOM (United States European Command) — The Combatant Command to which USAREUR-AF is the ASCC.

REFERENCES AND RELATED PUBLICATIONS

ADP 3-13, Information — Establishes information as combat power and the foundation for information advantage. This publication implements ADP 3-13 at the senior leader level.

ADP 5-0, The Operations Process — Defines the MDMP and the role of information in planning and execution.

Data Literacy Technical Reference, Data Literacy — The technical companion to this publication. Contains procedural guidance, analytical methods, data lifecycle doctrine, and the USAREUR-AF 5-Layer Data Stack architecture. Senior leaders requiring technical depth should consult Data Literacy Technical Reference.

DoD Data Strategy (2020) — Establishes the VAUTI framework (Visible, Accessible, Understandable, Trustworthy, Interoperable) as the standard for enterprise data quality and use. Referenced in this publication for the VAUTI decision framework and the principle of decision dominance.

Army CIO Guidance (2024) and Unified Data Reference Architecture (UDRA) v1.1 (2025) — The Army Data Plan (2022) established the foundational framework; current governance and architecture are codified in subsequent Army CIO guidance (2024) and the Unified Data Reference Architecture (UDRA) v1.1 (2025). The C2DAO is responsible for ensuring USAREUR-AF compliance with these documents.

NATO Architecture Framework version 4 (NAFv4) — The NATO standard for data and system architecture descriptions. USAREUR-AF data operations must align to NAFv4 for coalition interoperability.

Allied Joint Publications (AJP-01, AJP-3, AJP-5) — NATO joint doctrine governing operations, conduct, and planning. USAREUR-AF data products and analytical processes must support interoperability with these publications.

USAREUR-AF G6/Data — The authoritative source for technical architecture guidance, ontology standards, NATO NAFv4 alignment resources, and the USAREUR-AF 5-Layer Data Stack implementation documentation. Senior leaders seeking to understand the technical architecture underpinning USAREUR-AF data operations should direct their data staff to coordinate with G6/Data.

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